Railway Reservation System

RAILNEXUS

Reservations • Queue promotion • Announcements
Boarding • Reports • Audit

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Advanced Database Management Systems • October 2026

Project Scope and Five Actors

Passenger

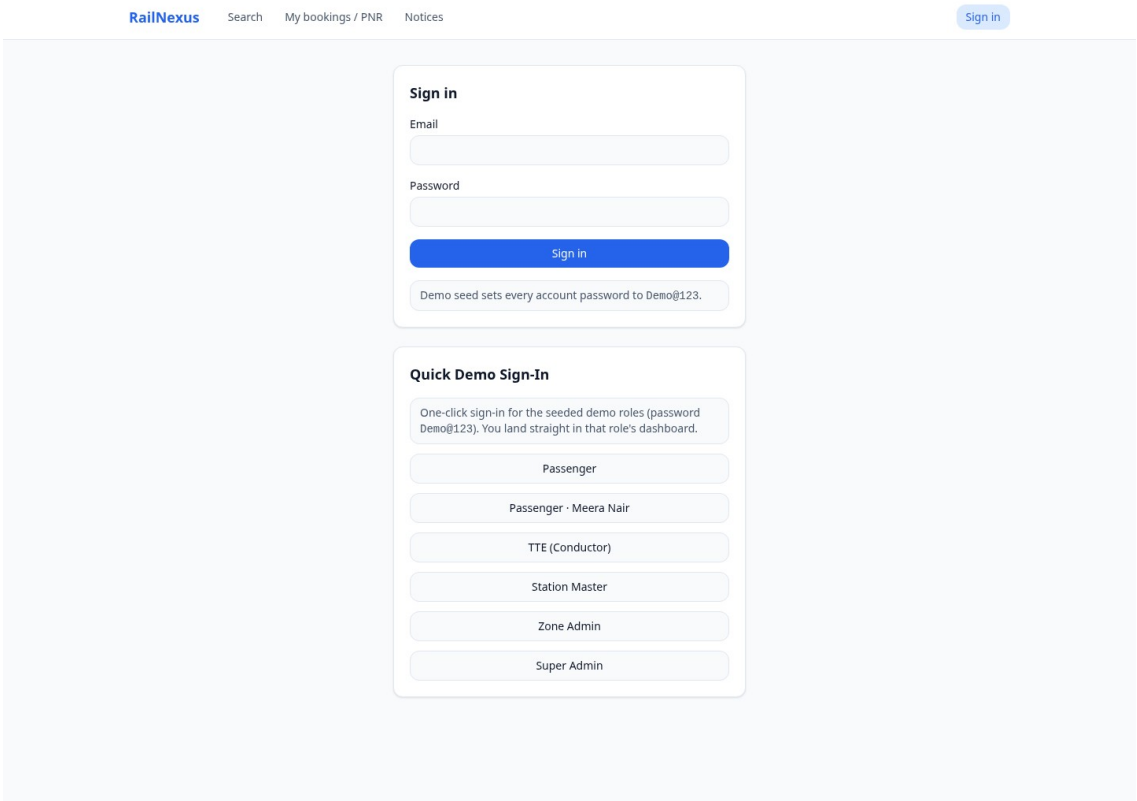
Search, book, pay, cancel and view notices.

Operational staff

TTE records boarding; Station Master publishes announcements.

Administration

Zone Admin sees assigned-zone reports; Super Admin sees all zones and audit events.



Three Database Technologies

Oracle

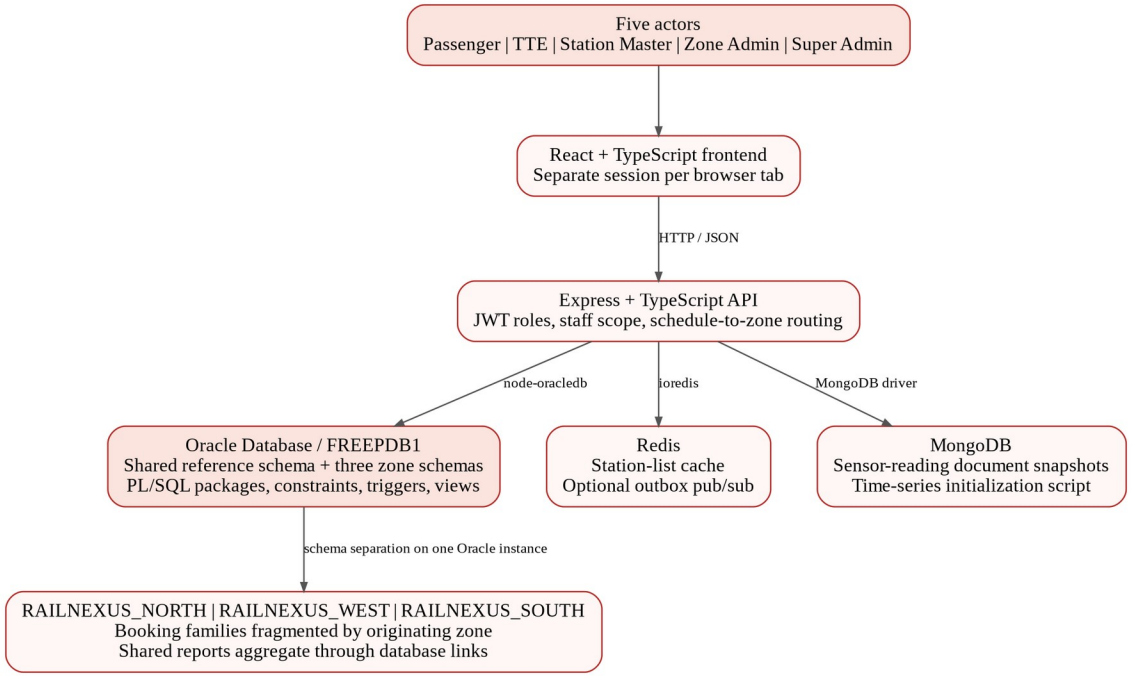
Bookings, tickets, payments, announcements, triggers and reports.

Redis

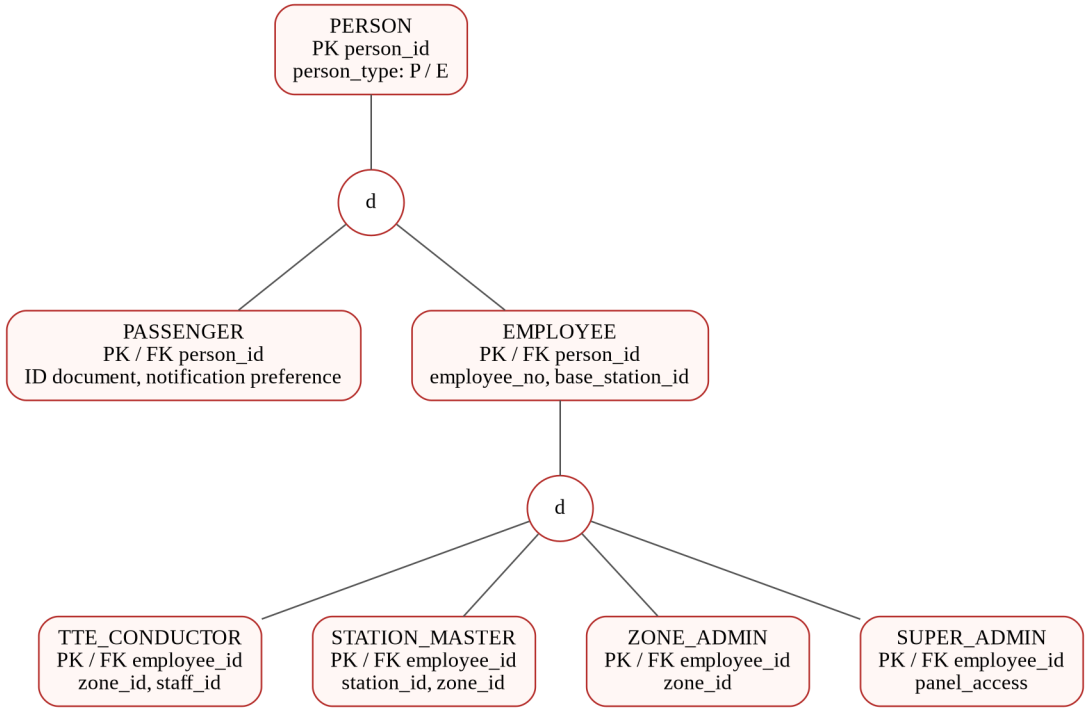
Fast station-list caching; optional outbox pub/sub.

MongoDB

Best-effort sensor-reading document snapshots.



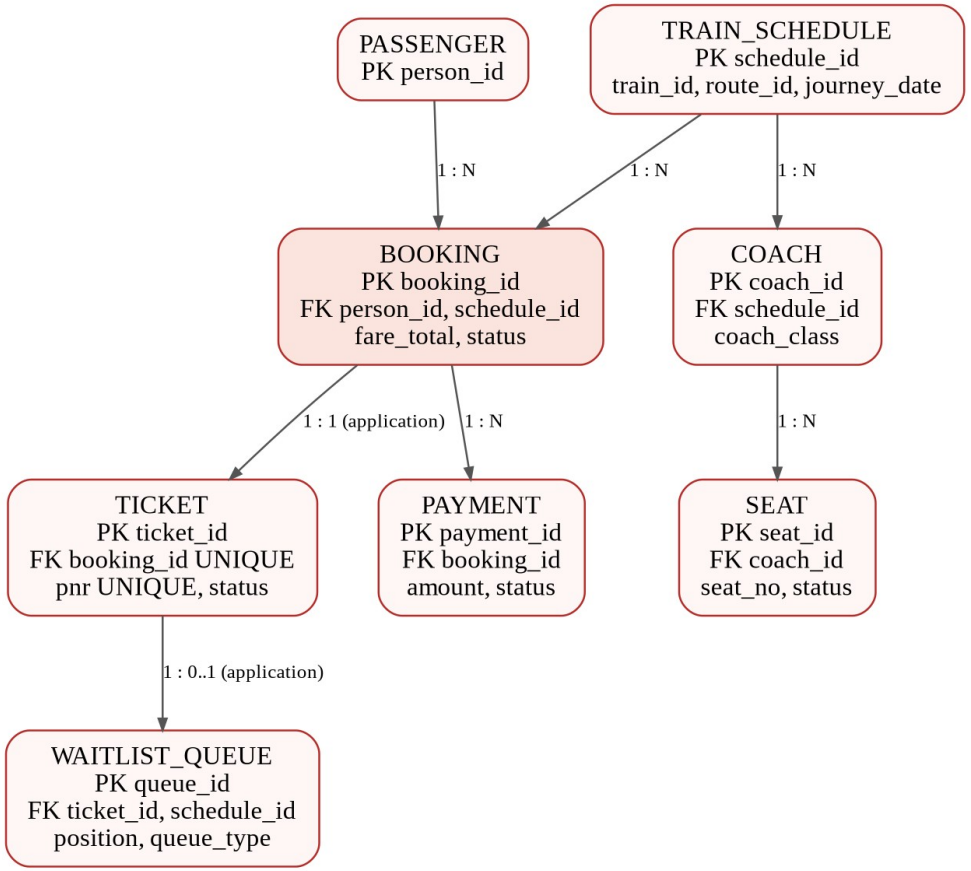
EER: Identity and Role Specialization



PERSON → Passenger / Employee; Employee → four staff roles.

d = disjoint. Subtype primary keys also reference their supertype.

EER: Reservation Relationships



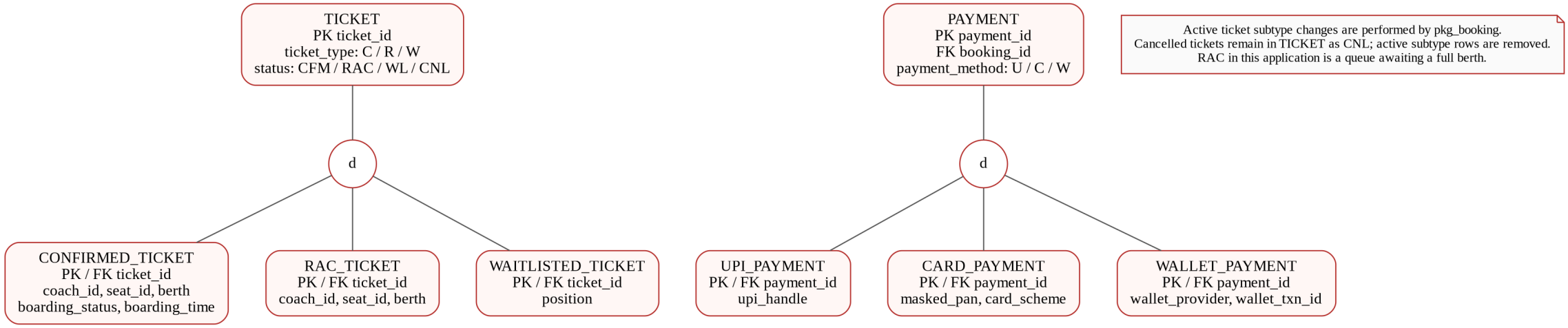
One passenger → many bookings

One journey → coaches → seats

One booking → one ticket

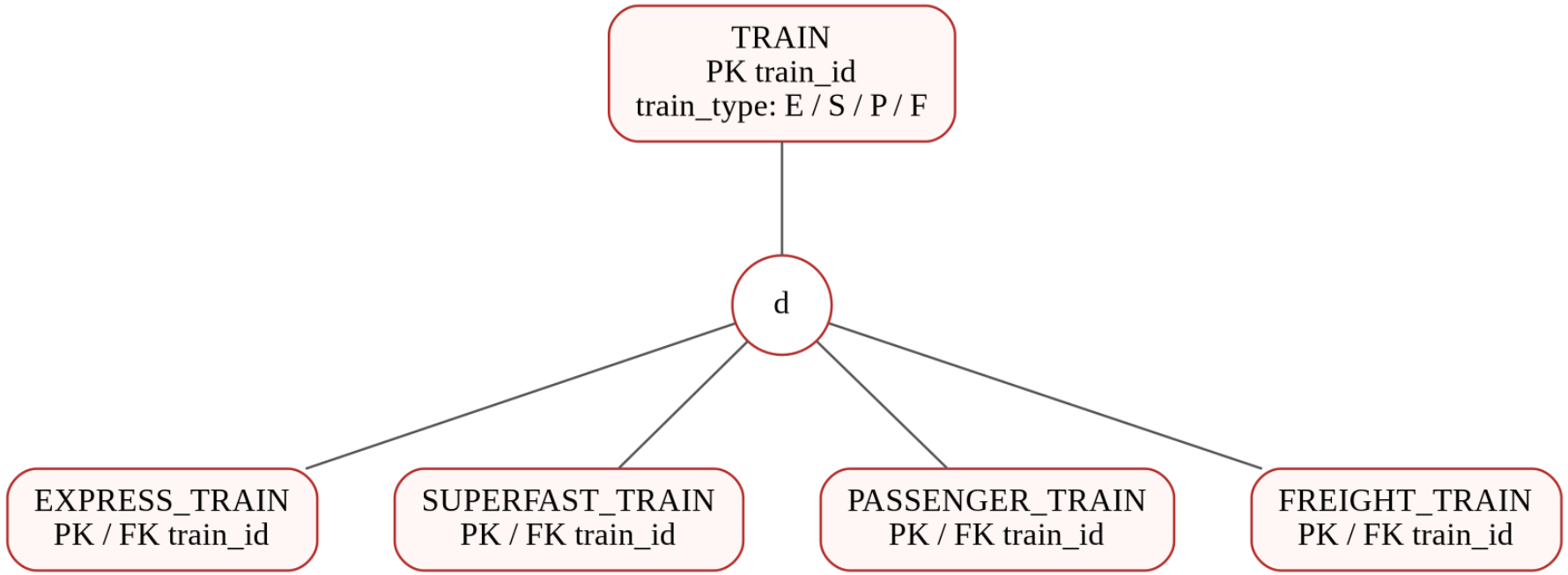
Payments and queue entries reference the reservation lifecycle.

EER: Ticket and Payment Specialization



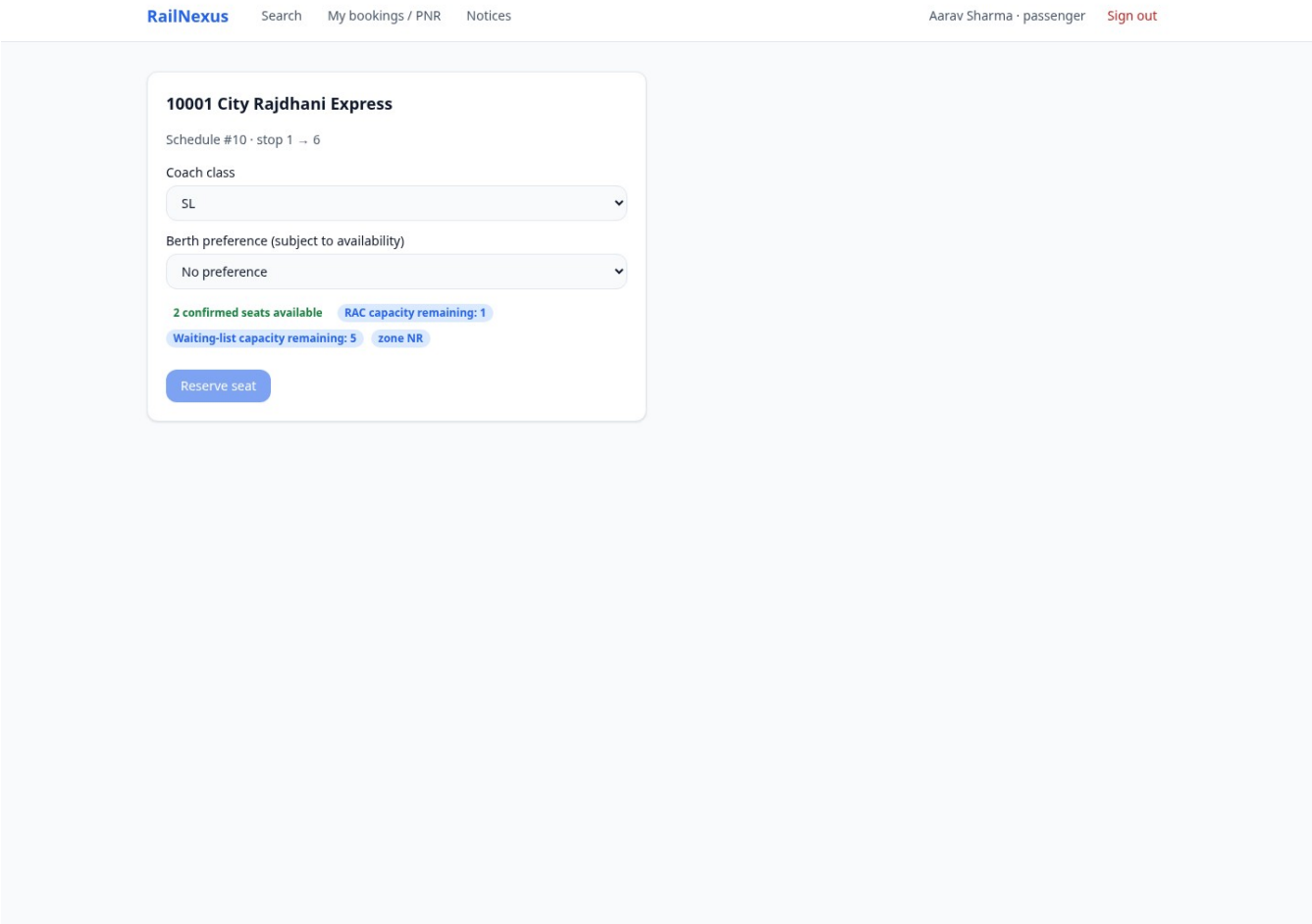
Common identity remains in the supertype; status- or method-specific fields remain in detail tables.

EER: Train Specialization



TRAIN → Express / Superfast / Passenger / Freight. Passenger search excludes freight services.

Search and Limited Inventory



Regular journeys

1–30 October 2026.

Per class

2 confirmed seats, 1 RAC place, 5 WL places.

Normal frontend

Search → Book → My bookings.

PL/SQL: Booking and Fare Calculation

Booking procedure

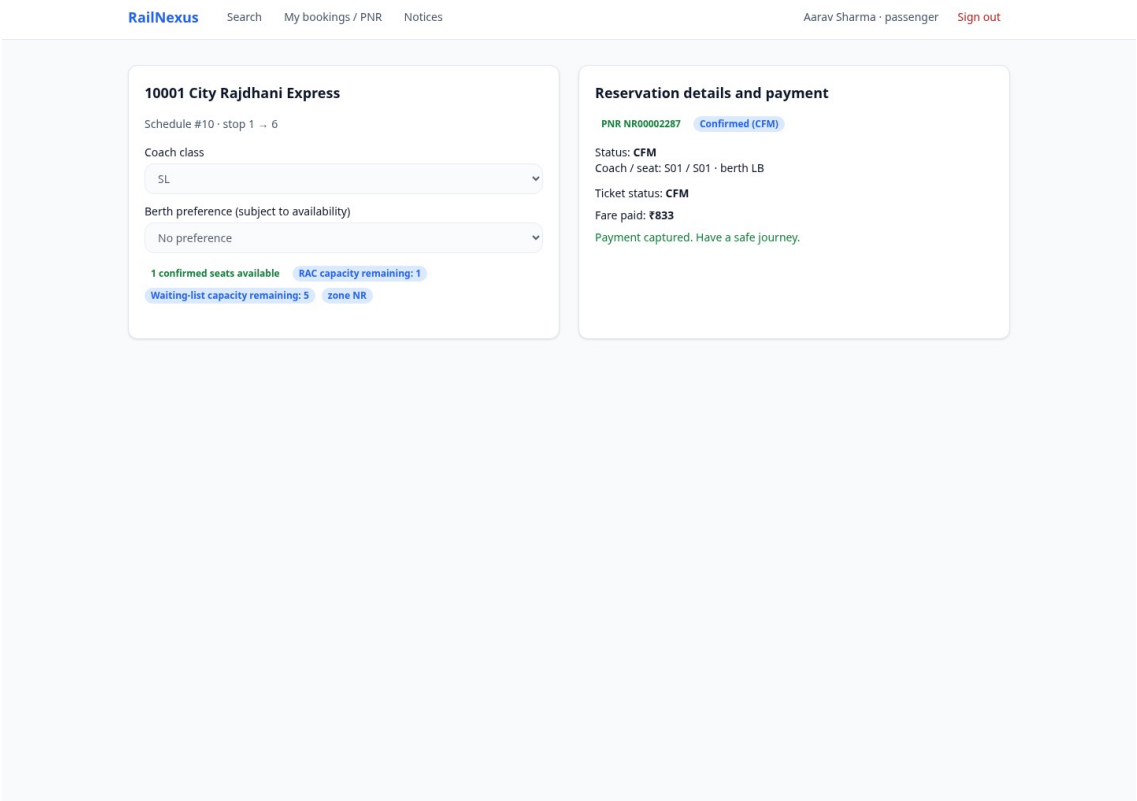
pkg_booking.book_ticket validates, locks inventory and allocates CFM / RAC / WL.

Fare function

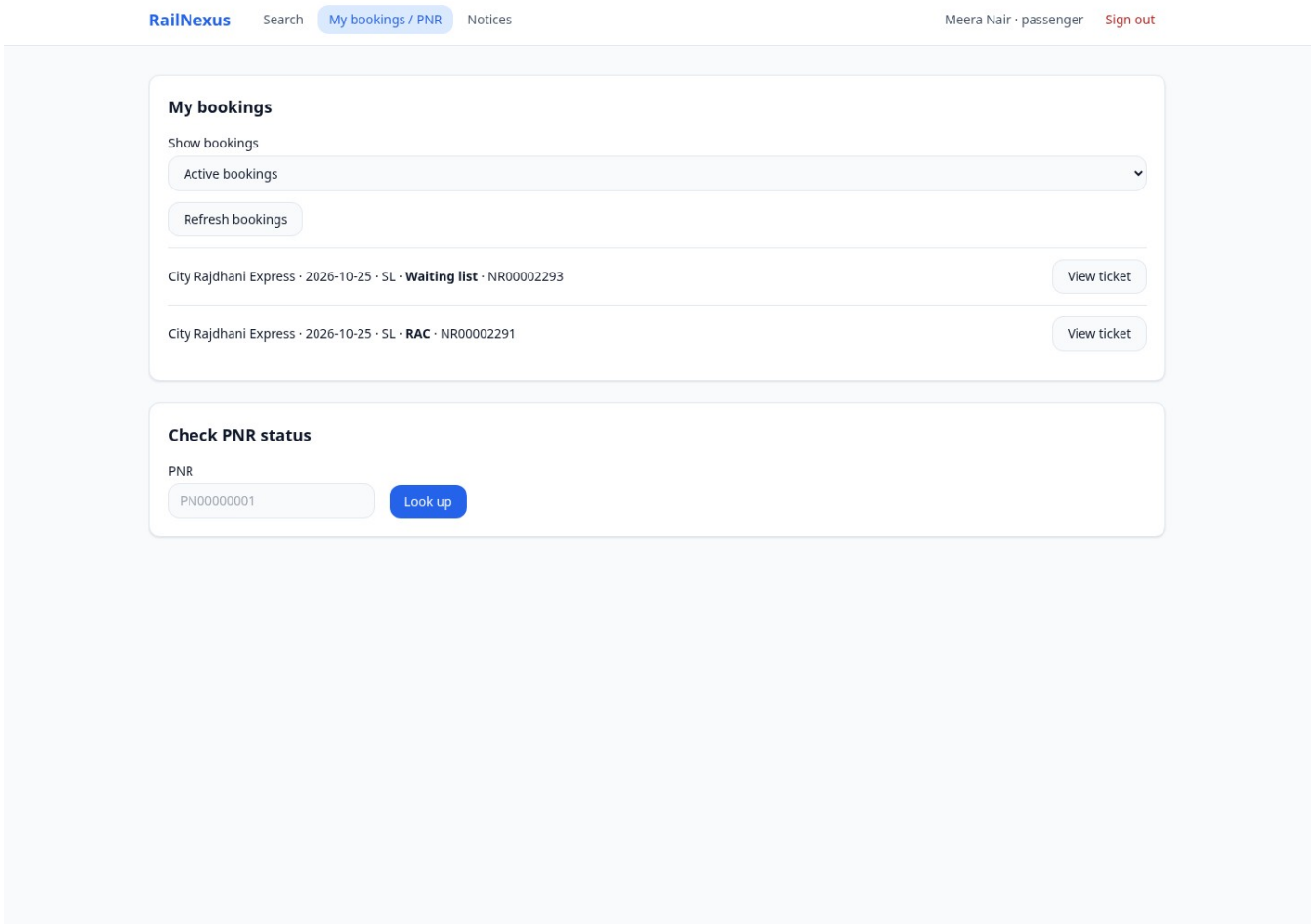
pkg_fare.compute_fare uses distance, class and the effective fare rule.

Request safety

Unique request IDs support repeated-call handling.



Two Passengers: Before Cancellation



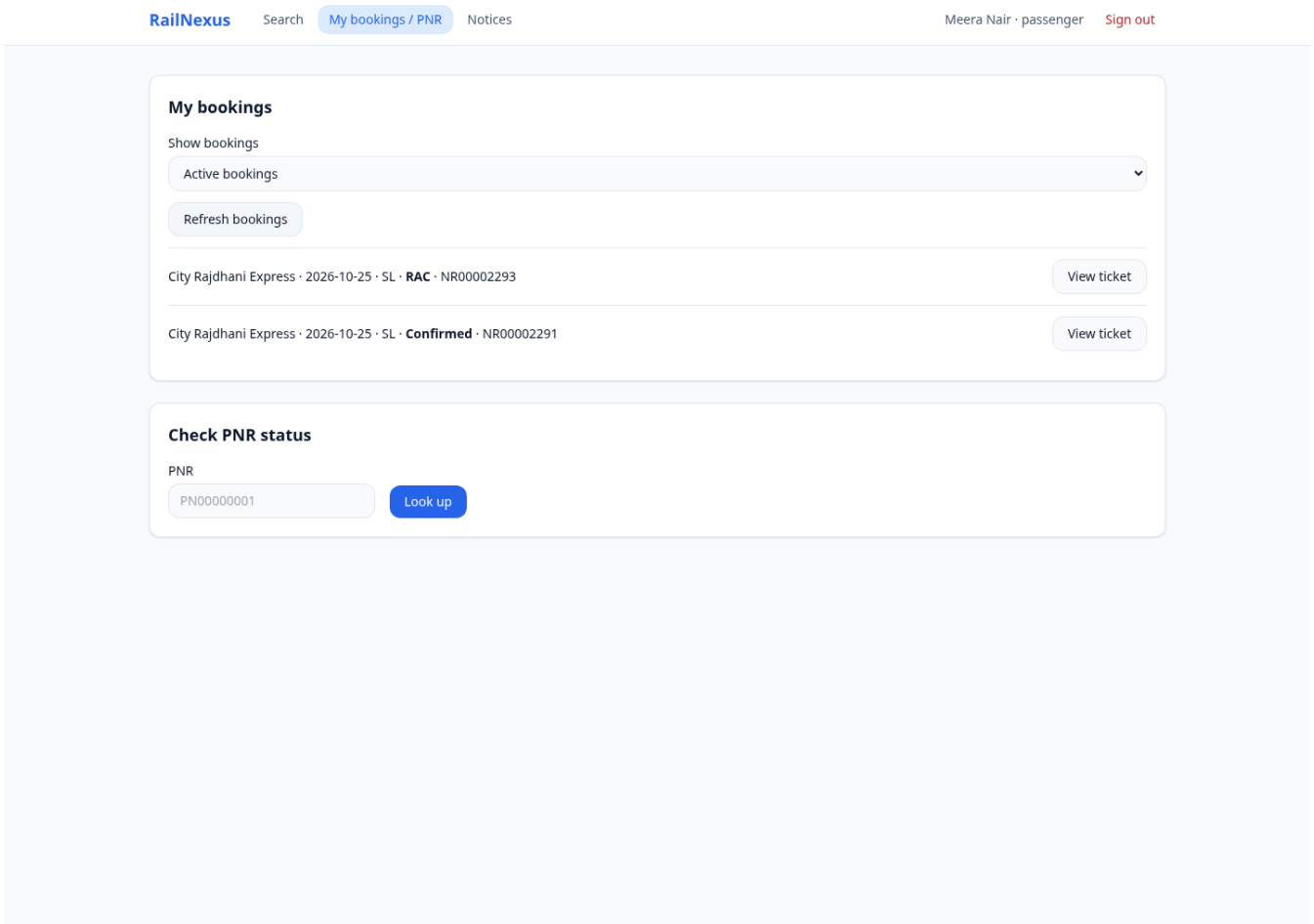
Same train, date and class

Aarav: 2 Confirmed

Meera: RAC + WL

No confirmed seats remain.

Cancellation: Automatic Queue Promotion



Aarav cancels one CFM ticket.

Meera: RAC → CFM

Meera: WL → RAC

pkg_booking.cancel_ticket updates allocation, queue and audit in a transaction.

Payment, Refund and Integrity

Payment

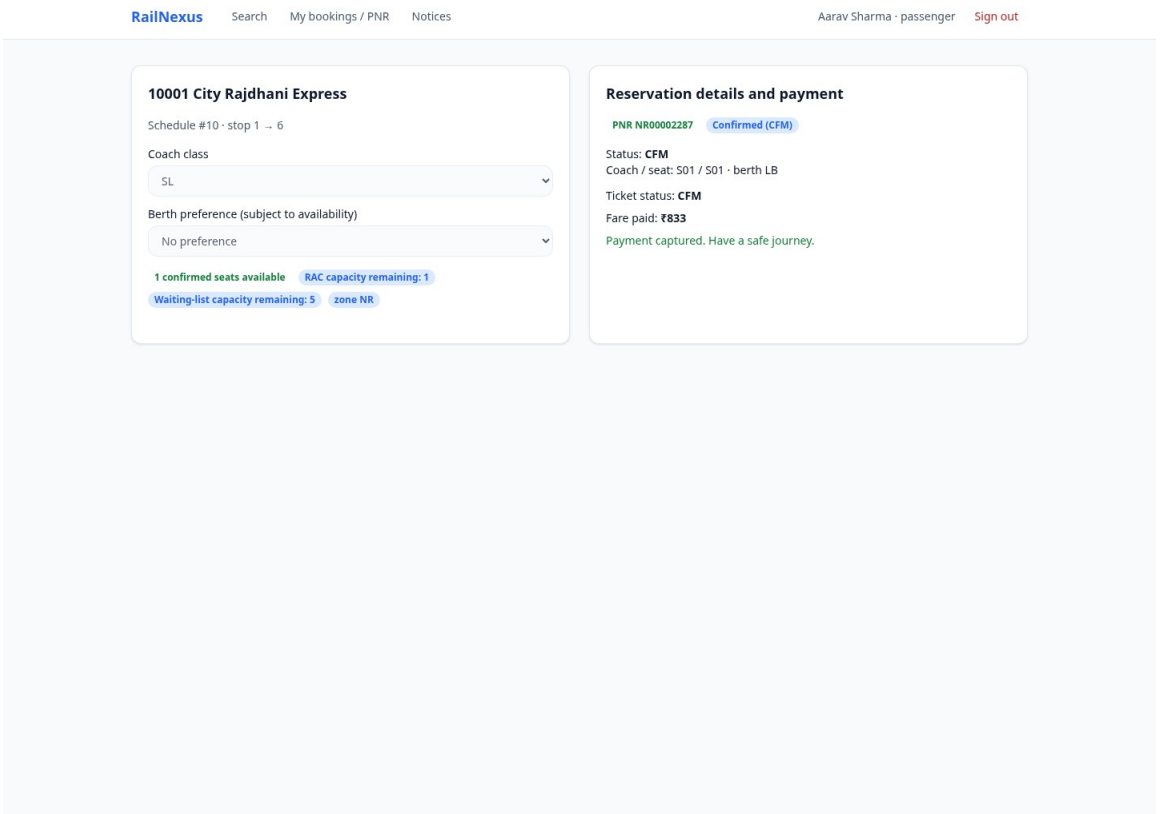
UPI / Card / Wallet capture is simulated; fare is validated on the server.

Refund

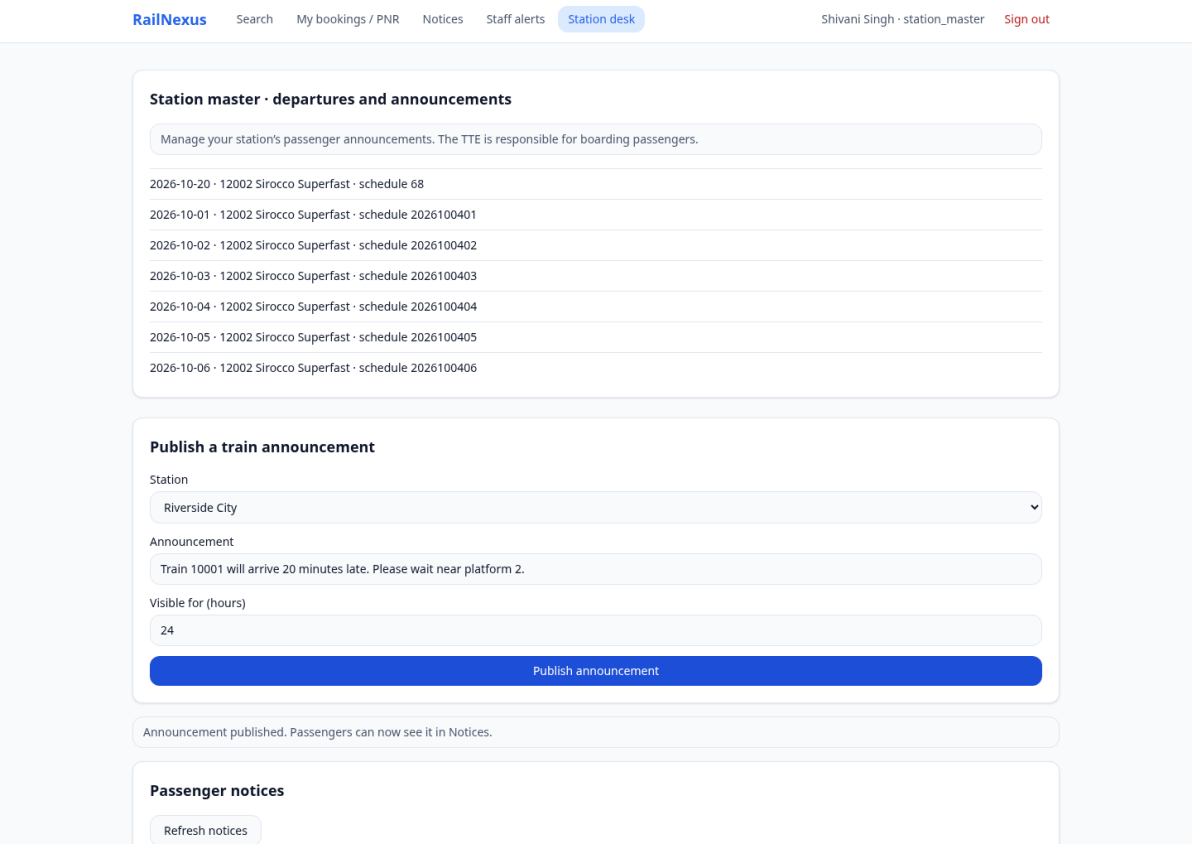
Cancellation requests the captured refundable amount.

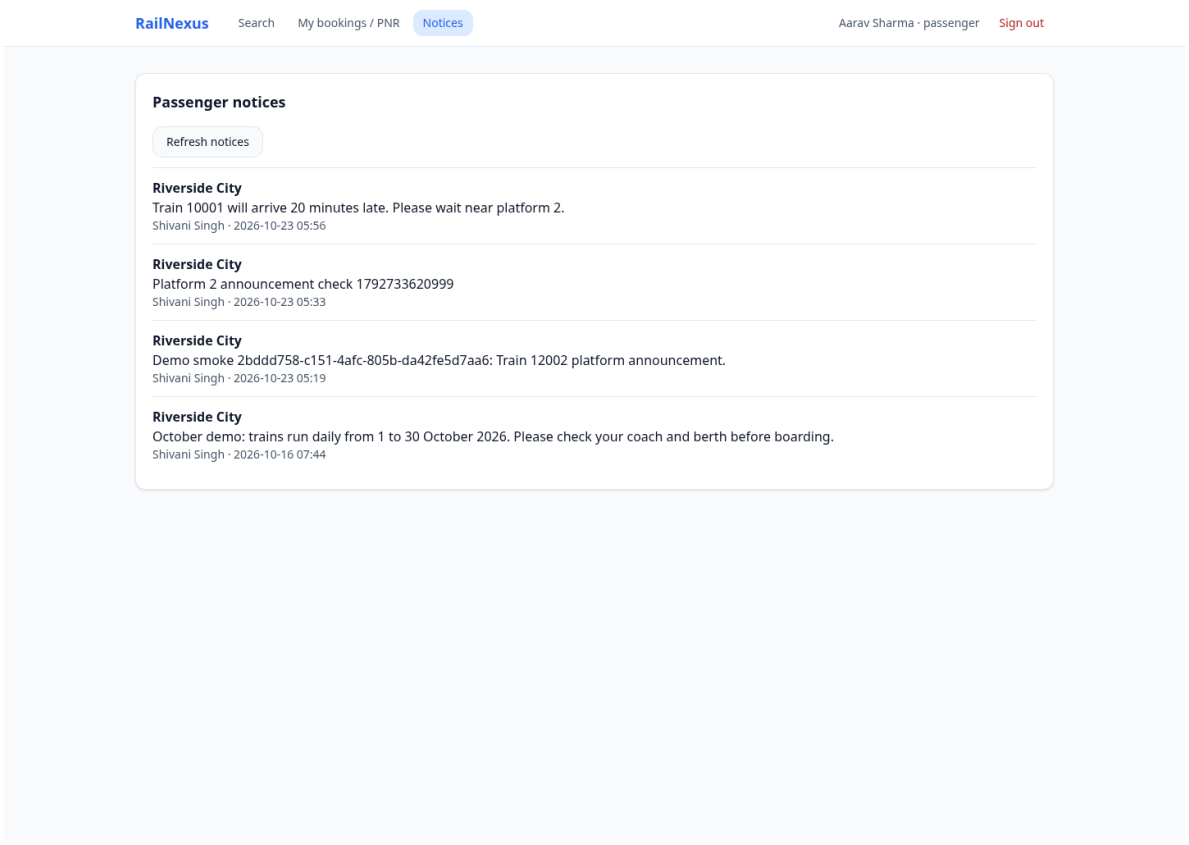
Integrity

Keys, status checks, ownership and request identifiers protect records.



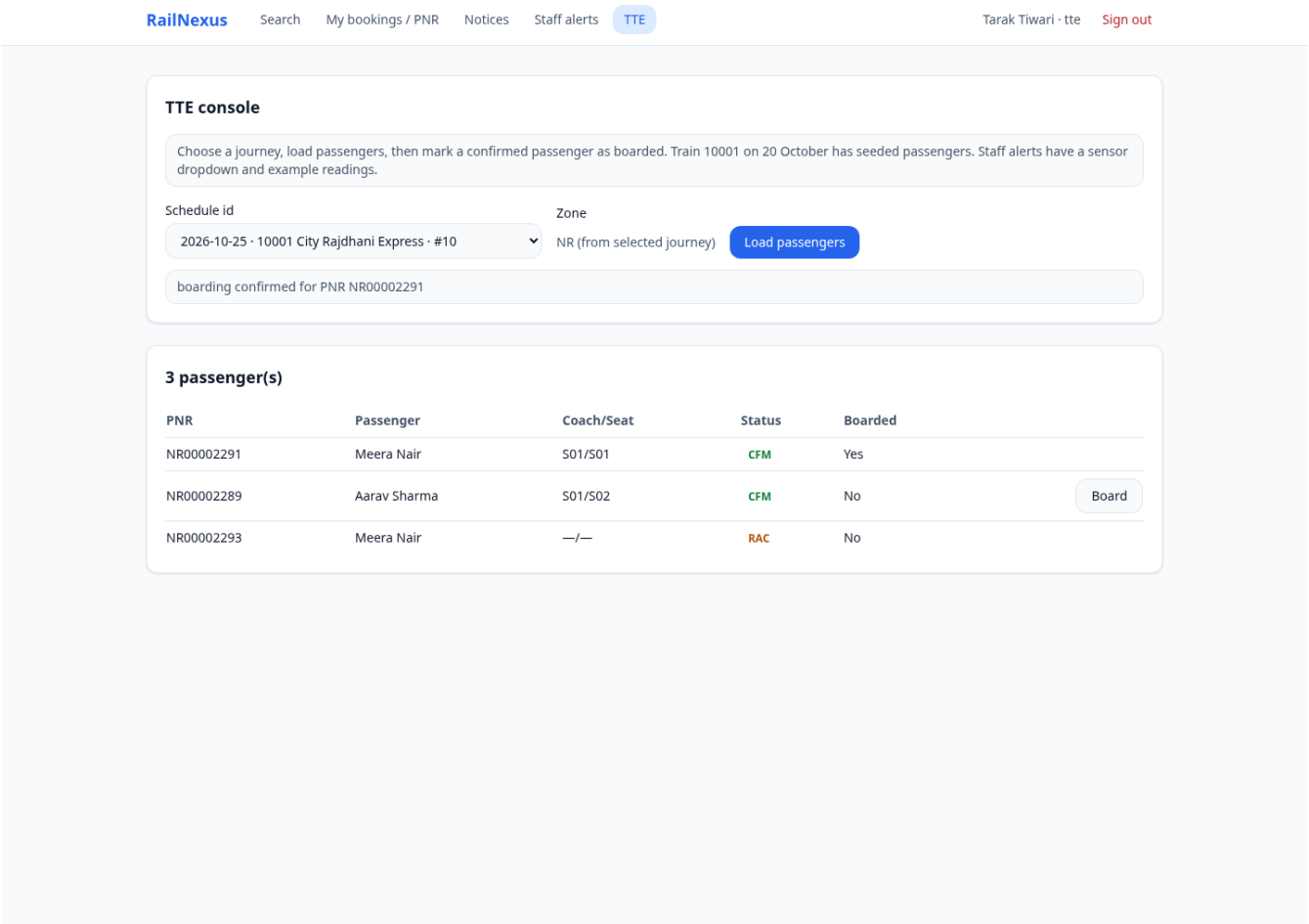
Station Master → Passenger Notice





Publish at the assigned station → refresh Passenger Notices → the same message appears.

Passenger Booking → TTE Boarding



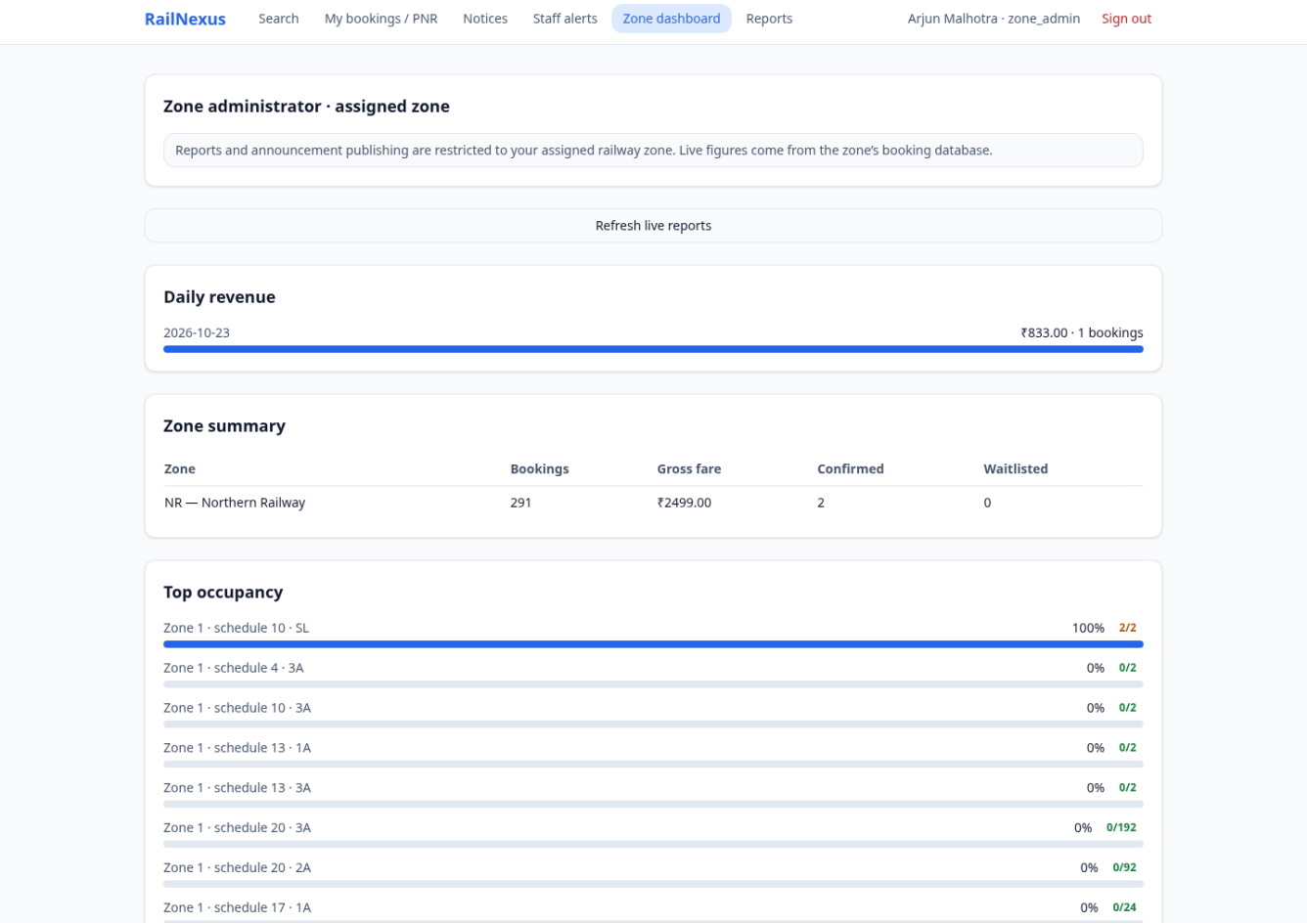
Select the same journey.

Load the passenger list.

Mark a CFM passenger as boarded.

pkg_tte.board_ticket records the update.

Zone and Super Administrator Reports



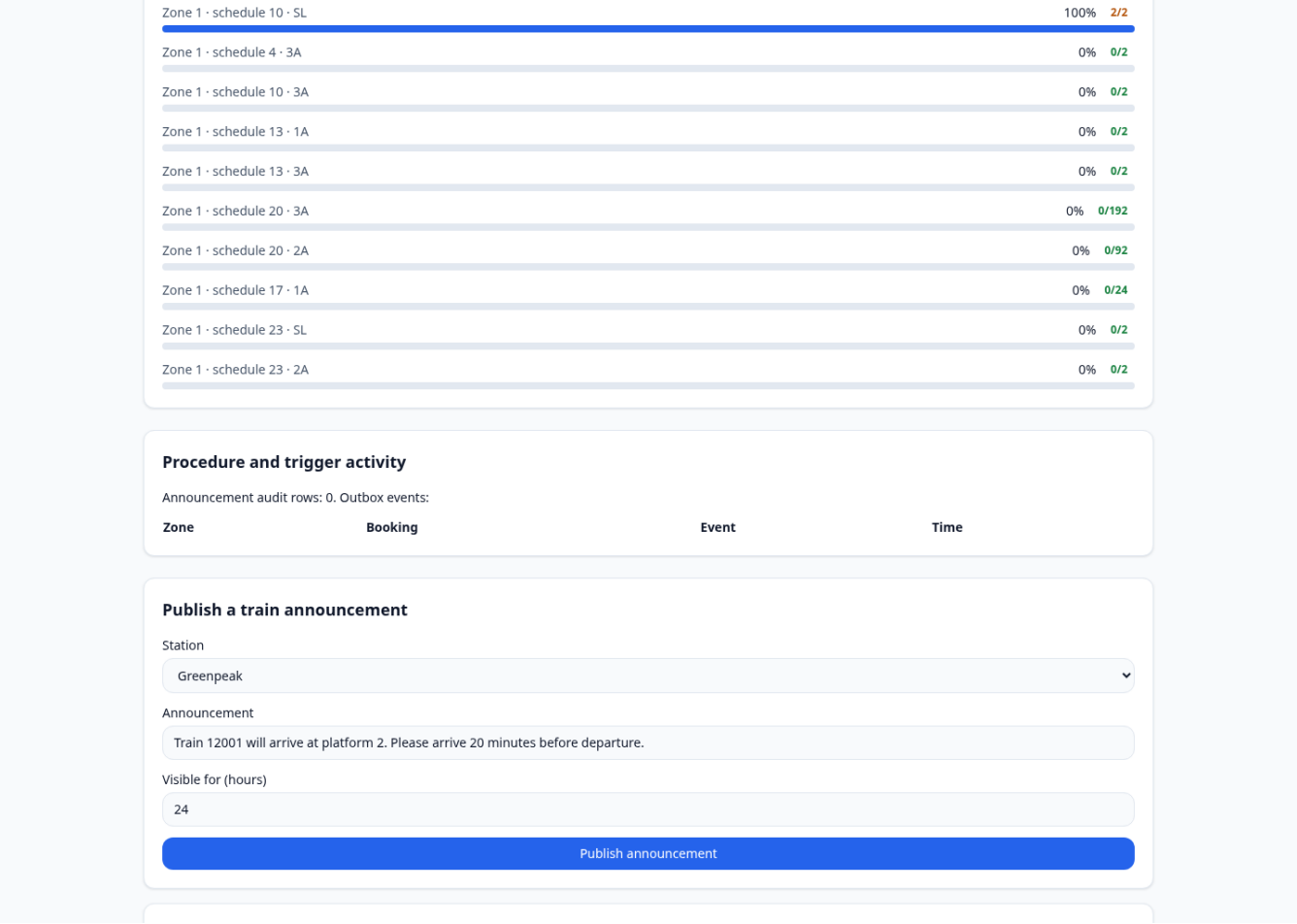
Zone Admin: assigned zone only.

Super Admin: all three zones.

Views show live aggregates.

Materialized reporting data is refreshed on demand.

Triggers and Audit Evidence



Package inserts booking audit.

trg_audit_to_outbox inserts an outbox event.

Announcement trigger records publication.

Queue promotion itself is procedure logic.

ADBMS Concepts and Source Files

Procedures / functions	db/plsql/pkg_booking.sql; pkg_fare.sql
Triggers	db/plsql/outbox_audit_trigger.sql; pkg_announcements.sql
EER and constraints	db/shared_ddl/01_core_schema.sql; db/zone_common/booking_family.sql
Fragmentation / views	db/distributed/db_links.sql; db/reporting/report_views.sql
Object types / JSON	db/or/route_journey_types.sql; db/json/pkg_json_rules.sql

Validation and Project Boundaries

Verified

276 real API assertions; 31 backend + 2 frontend unit tests; production builds passed.

Connected workflows

Two-passenger promotion, staff publication, TTE boarding and role restrictions.

Prototype limits

Simulated payments; one Oracle instance with three zone schemas; refresh-based updates.

Conclusion

One actor changes data.

Another actor sees the database result.

Booking → CFM / RAC / WL

Cancellation → promotion and refund

Announcement → Passenger Notices

Boarding → updated TTE record

Transactions → consistent records and audit